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MAT 213 Brief Calculus

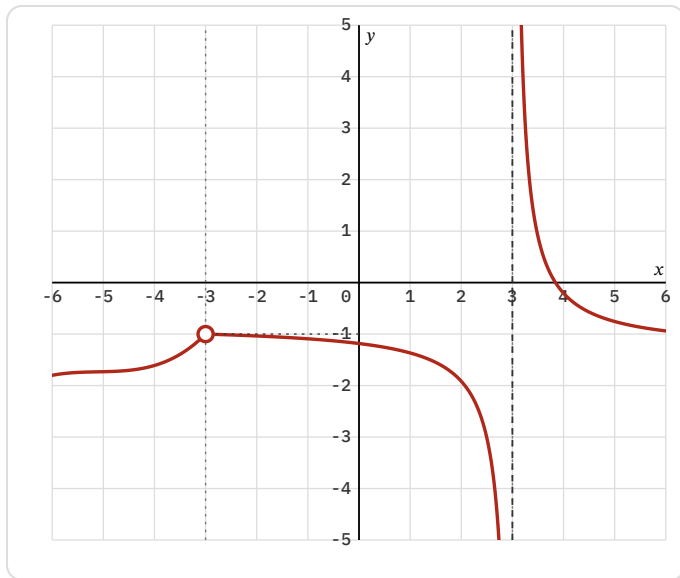
Mock Midterm A — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 58 points

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE.



a.) $\lim_{x \rightarrow -3^-} f(x)$ (2)

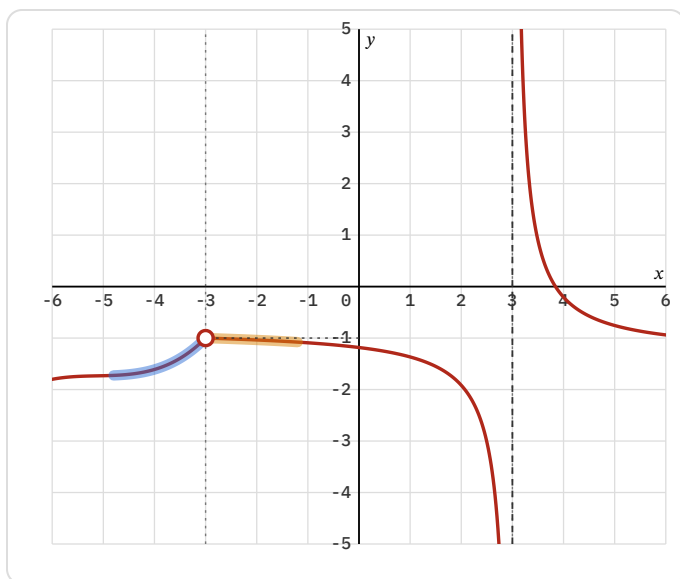
-1

b.) $f(3)$ (2)

DNE

4 pts

WORKED SOLUTION



— approaching $x = -3$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

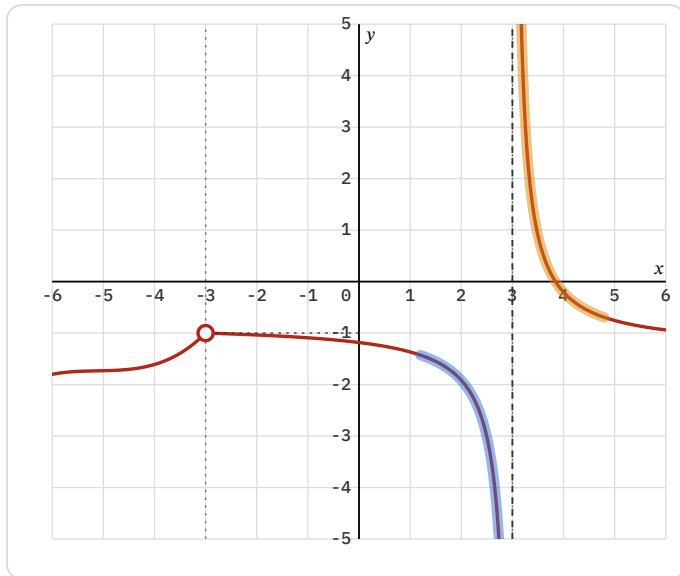
At $x = -3$

There is no filled dot on the line $x = -3$, so $f(-3)$ is undefined $\rightarrow$ DNE.

Tracing the curve toward $x = -3$ from the left, the heights approach -1 (open or closed dot doesn't matter).

Tracing the curve toward $x = -3$ from the right, the heights approach -1 (open or closed dot doesn't matter).

Both one-sided limits equal -1, so $\lim_{x \rightarrow -3} f(x) = -1$ — even though $f(-3)$ is undefined. The limit ignores the value AT the point.



— approaching $x = 3$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

At $x = 3$

There is no filled dot on the line $x = 3$, so $f(3)$ is undefined $\rightarrow$ DNE.

From the left, the curve runs along the asymptote $x = 3$ downward, so the limit is $-\infty$.

From the right, the curve runs along the asymptote $x = 3$ upward, so the limit is ∞ .

The left and right limits don't match, so $\lim_{x \rightarrow 3} f(x)$ DNE.

2. Let $f(x) = \frac{x^3+8}{x+2}$. Use the table to see what the outputs do as x approaches -2 .

3 pts

x	-2.01	-2.001	-2.0001	-2	-1.9999	-1.999	-1.99
$f(x)$	12.0601	12.006001	12.0006	?	11.9994	11.994001	11.9401

- a.) Can you find $\lim_{x \rightarrow -2} f(x)$ just by plugging in $x = -2$? (1)

(B) No — $x = -2$ makes a denominator 0, so $f(-2)$ is undefined

- b.) $\lim_{x \rightarrow -2} f(x) =$ (2)

12

WORKED SOLUTION

Reading the left three columns, the outputs head toward **12**; the right three head toward **12**.

Both sides agree, so the limit is **12**.

$$\lim_{x \rightarrow -2} \frac{x^3 + 8}{x + 2} = 12$$

Why: $\frac{(x+2)(x^2-2x+4)}{(x+2)} = x^2 - 2x + 4 \rightarrow 12$

3.

Suppose

4 pts

$$\lim_{x \rightarrow 4} f(x) = 1, \quad \lim_{x \rightarrow 4} g(x) = -4, \quad \lim_{x \rightarrow 4} h(x) = 25$$

Use the properties of limits to find each limit.

a.) $\lim_{x \rightarrow 4} ([f(x)]^3 - g(x))$ (2)

5

b.) $\lim_{x \rightarrow 4} x^2 f(x)$ (2)

16

WORKED SOLUTION

a. $\lim_{x \rightarrow 4} ([f(x)]^3 - g(x)) = (1)^3 - (-4) = 1 + 4 = 5$

b. $\lim_{x \rightarrow 4} \sqrt{h(x)} f(x) = \sqrt{25} \cdot (1) = 5 \cdot (1) = 5$

c. $\lim_{x \rightarrow 4} \frac{g(x)-2}{f(x)} = \frac{-4-2}{1} = -6$

d. $\lim_{x \rightarrow 4} x^2 f(x) = (4)^2 \cdot (1) = 16$

The quotient property works here because the bottom limit isn't 0.

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{x-25}$ (2)

$\frac{1}{10}$

b.) $\lim_{h \rightarrow 0} \frac{\frac{1}{4+h} - \frac{1}{4}}{h}$ (2)

$-\frac{1}{16}$

4 pts

WORKED SOLUTION

a.

$$\lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{x-25} = \lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{(\sqrt{x}-5)(\sqrt{x}+5)} = \lim_{x \rightarrow 25} \frac{1}{\sqrt{x}+5} = \frac{1}{5+5} = \frac{1}{10}$$

b.

$$\lim_{h \rightarrow 0} \frac{\frac{1}{4+h} - \frac{1}{4}}{h} = \lim_{h \rightarrow 0} \frac{\frac{4-(4+h)}{4(4+h)}}{h} = \lim_{h \rightarrow 0} \frac{-h}{4(4+h)h} = \lim_{h \rightarrow 0} \frac{-1}{4(4+h)} = -\frac{1}{16}$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results.

4 pts

a.) $\lim_{x \rightarrow \infty} \frac{-3x^3 - x^2}{5x}$ (2)

$-\infty$

b.) $\lim_{x \rightarrow \infty} \frac{3x^3}{-5x^4 - 5x^3 + 7x^2}$ (2)

0

WORKED SOLUTION

a. Divide every term by x , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-3x^3 - x^2}{5x} = \lim_{x \rightarrow \infty} \frac{-3x^2 - x}{5}$$

The bottom $\rightarrow 5$, but the top is dominated by $-3x^2$, which $\rightarrow -\infty$ as $x \rightarrow \infty$; dividing by 5 gives $-\infty$.

b. Divide every term by x , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{7x + 2}{-5x - 5} = \lim_{x \rightarrow \infty} \frac{7 + \frac{2}{x}}{-5 - \frac{5}{x}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{7}{-5} = -\frac{7}{5}$.

c. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{3x^3}{-5x^4 - 5x^3 + 7x^2} = \lim_{x \rightarrow \infty} \frac{\frac{3}{x}}{-5 - \frac{5}{x} + \frac{7}{x^2}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow -5$, so the limit is 0.

6.

Let

4 pts

$$G(x) = \begin{cases} x^2 - 2 & x < -2 \\ -x & x > -2 \end{cases}$$

Find each quantity. If it does not exist, write DNE.

a.) $G(-2)$ (2)

DNE

b.) $\lim_{x \rightarrow \infty} G(x)$ (2)

 $-\infty$ **WORKED SOLUTION** **$G(-2)$:** the pieces cover $x < -2$ and $x > -2$ only, so $G(-2)$ is undefined $\rightarrow$ DNE.**Left:** use $x^2 - 2$: $(-2)^2 - 2 = 2$.**Right:** use $-x$: $-1(-2) + 0 = 2$.**Two-sided:** both sides give 2, so the limit is 2 (even though $G(-2)$ is undefined). **$G(1)$:** $1 > -2$, so use $-x$: $-1(1) + 0 = -1$.**End behavior:** as $x \rightarrow -\infty$ we're on the $x^2 - 2$ piece — an even-degree term with positive leading coefficient $\rightarrow \infty$. As $x \rightarrow \infty$ we're on the line with slope -1 $\rightarrow -\infty$.

7. Let

4 pts

$$f(x) = \begin{cases} \frac{2x^2 - 5x + 3}{2x - 3} & x < \frac{3}{2} \\ 4x + 4 & x \geq \frac{3}{2} \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 5} f(x)$ (2)

24

b.) $\lim_{x \rightarrow \frac{3}{2}} f(x)$ (2)

DNE

WORKED SOLUTION

For $x < \frac{3}{2}$, factor the top: $\frac{(2x-3)(x-1)}{2x-3} = x - 1$ (valid since $x \neq \frac{3}{2}$ there).

a. $5 > \frac{3}{2} \rightarrow$ bottom piece: $4(5) + 4 = 24$.

b. $-2 < \frac{3}{2} \rightarrow$ top piece, simplified: $-2 - 1 = -3$.

c. The bottom piece includes $x = \frac{3}{2}$ ($\geq$), so $f(\frac{3}{2}) = 4(\frac{3}{2}) + 4 = 10$.

d. Left: $\lim_{x \rightarrow \frac{3}{2}^-} (x - 1) = \frac{1}{2}$.

e. Right: $\lim_{x \rightarrow \frac{3}{2}^+} (4x + 4) = 10$.

f. $\frac{1}{2} \neq 10$, so the limit DNE.

8. Is $f(x)$ continuous at $x = -1$?

3 pts

$$f(x) = \begin{cases} \frac{x^2 + 6x + 5}{x + 1} & x \neq -1 \\ 4 & x = -1 \end{cases}$$

a.) $f(-1)$ (2)

4

b.) Is f continuous at $x = -1$? (1)

(A) Yes — $f(-1)$ exists, the limit exists, and they are equal

WORKED SOLUTION

1. $f(-1) = 4$ (given directly by the second piece).

2. $\lim_{x \rightarrow -1} \frac{(x+1)(x+5)}{(x+1)} = \lim_{x \rightarrow -1} (x+5) = 4.$

3. $4 = 4$. All three conditions hold $\rightarrow$ **continuous**.

9. Classify each function at the given x -value.

2 pts

a.) $f(x) = x^2 + 5x - 4$ at $x = -1$ (1)

(A) Continuous

b.) $f(x) = \frac{x+2}{x^2+x-2}$ at $x = -2$ (1)

(B) Removable discontinuity

WORKED SOLUTION

a. Polynomials are continuous everywhere.

b. $0/0$ at -2 , but it cancels to $x - 2$, so the limit exists (-4) while $f(-2)$ is undefined $\rightarrow$ removable.

c. Factor: $\frac{(x+2)}{(x+2)(x-1)}$. The $(x+2)$ cancels, leaving $\frac{1}{x-1} \rightarrow -\frac{1}{3} \rightarrow$ removable.

10.

Let $C(q) = q^2 + 3q + 200$, where $C(q)$ represents the cost, in dollars, of producing q items. Find the average rate of change over $[4, 8]$.

a.) $C(4)$ (2)

228

b.) Which sentence correctly states the result **with units**? (1)

(B) Between $q = 4$ and $q = 8$ items, the cost changes at an average rate of 15 dollars per item.

3 pts

WORKED SOLUTION

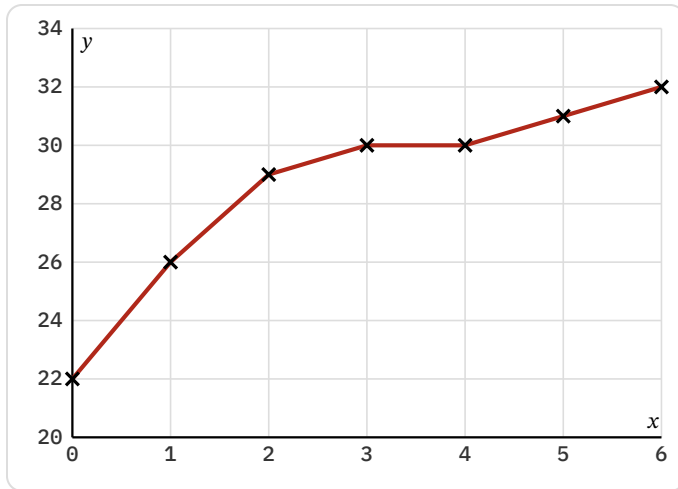
$$C(4) = 228, \quad C(8) = 288$$

$$\frac{C(8) - C(4)}{8 - 4} = \frac{288 - (228)}{4} = \frac{60}{4} = 15 \text{ dollars per item}$$

It's an *average* over an interval (not an instant), and the units are dollars per item — dollars per item. The total change would be 60 dollars.

11.

The graph shows the closing price (in dollars) of a stock at the start of each month in 2026, where $x = 0$ is January, $x = 1$ is February, and so on.



a.) Average rate of change on $[2, 5]$ (dollars per month) (2)

$$\frac{2}{3}$$

b.) On which interval is the average rate of change greatest? (1)

(A) $[0, 2]$

3 pts

WORKED SOLUTION

$$\frac{f(5) - f(2)}{5 - 2} = \frac{31 - 29}{3} = \frac{2}{3} \text{ dollars per month}$$

Between Mar and Jun the price changed by $\frac{2}{3}$ dollars per month on average — the slope of the secant line joining those two points.

Greatest-rate check: $[0, 2]: \frac{7}{2}$, $[2, 4]: \frac{1}{2}$, $[4, 6]: 1 \rightarrow [0, 2]$.

12.

Let $f(x) = 2x^2 + 2x + 3$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) (3)

$$4x + 2h + 2$$

b.) As $h \rightarrow 0$, what value do these approach? (2)

$$14$$

5 pts

WORKED SOLUTION

$$f(x+h) = 2(x+h)^2 + 2(x+h) + 3 = 2x^2 + 4xh + 2h^2 + 2x + 2h + 3$$

$$f(x+h) - f(x) = 4xh + 2h^2 + 2h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{h(4x + 2h + 2)}{h} = 4x + 2h + 2$$

Now substitute:

$$x = 3, h = 1 : 16$$

$$x = 3, h = 0.1 : \frac{71}{5} = 14.2$$

$$x = 3, h = 0.01 : \frac{701}{50} = 14.02$$

Letting $h \rightarrow 0$ leaves **14**: the instantaneous rate of change (slope of the tangent line) at $x = 3$. That's exactly what section 1.4 formalizes.

13.

Let $f(x) = x^3 - x$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ (3)
 $3x^2 - 1$

b.) What does $f'(-2) = 11$ mean? (1)

(A) The instantaneous rate of change of f at $x = -2$ is 11 — the slope of the tangent line there.

4 pts

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{[(x+h)^3 - 1(x+h)] - (x^3 - x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - 1h}{h} = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 - 1) = 3x^2 - 1 \end{aligned}$$

Then $f'(-2) = 11$ and the slope of the tangent at $x = 2$ is $f'(2) = 11$.

$f'(-2)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14. Let $f(x) = -x^2 + x$. Find the equation of the tangent line to the graph of f at $x = -2$. 5 pts

a.) $f(-2)$ (2)
 -6

b.) Tangent line (3)
 $y = 5x + 4$

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[-(x+h)^2 + (x+h)] - (-x^2 + x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-x^2 - 2xh - h^2 + x + h + x^2 - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2xh - h^2 + h}{h} = \lim_{h \rightarrow 0} \frac{h(-2x - h + 1)}{h} \\ &= \lim_{h \rightarrow 0} (-2x - h + 1) = -2x + 1 \end{aligned}$$

Point: $f(-2) = -6$, so the point is $(-2, -6)$. Slope: $m = f'(-2) = 5$.

$$y + 6 = 5(x + 2) \Rightarrow y = 5x + 4$$

Answers in point-slope form are accepted too.

15.

Let $T(t) = -t^2 + 30t + 70$ be the temperature, in °F, of an oven t minutes after it is turned on.

a.) $T'(t) =$ (3)
 $-2t + 30$

b.) Interpret $T'(12) = 6$. (1)
(A) At $t = 12$ minutes, the oven's temperature is increasing at a rate of 6 degrees Fahrenheit per minute.

4 pts

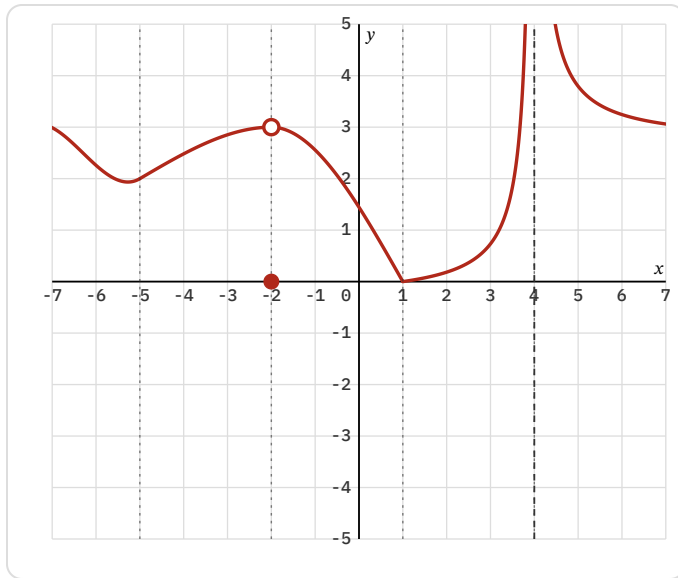
WORKED SOLUTION

$$\begin{aligned} T'(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[-(t+h)^2 + 30(t+h) + 70] - (-t^2 + 30t + 70)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-t^2 - 2th - h^2 + 30t + 30h + 70 + t^2 - 30t - 70}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2th - h^2 + 30h}{h} = \lim_{h \rightarrow 0} \frac{h(-2t - h + 30)}{h} \\ &= \lim_{h \rightarrow 0} (-2t - h + 30) = -2t + 30 \\ T'(12) &= -2(12) + 30 = 6 \end{aligned}$$

Positive/negative tells the direction, and the units are degrees Fahrenheit per minute: at that instant the oven's temperature is increasing at 6 degrees Fahrenheit per minute. (The value $T(12) = 286$ is a different quantity.)

16.

At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there.



a.) At $x = -5$ (1)

(A) Continuous and differentiable

b.) At $x = 4$ (1)

(C) Not continuous (so not differentiable)

2 pts

WORKED SOLUTION

$x = -5$: Smooth and unbroken — a single tangent line fits → continuous and differentiable.

$x = -2$: There's a hole (the dot isn't on the curve) → not continuous, so not differentiable.

$x = 1$: No break, but the curve comes to a sharp point (corner): the slopes from the left and right disagree → continuous, not differentiable.

$x = 4$: Vertical asymptote → not continuous, so not differentiable.
